

“Once”: Multi-perspective and Interactive VR Narrative Film

Zixiao Liu

School of New Media Art and Design,
Beihang University, Beijing, China
liuzixiao26@163.com

Lutong Li

School of New Media Art and Design,
Beihang University, Beijing, China
1531520585@qq.com

Shuo Yan

School of New Media Art and Design,
State Key Laboratory of Virtual
Reality Technology and Systems,
Beihang University, Beijing, China
shuoyan@buaa.edu.cn



Figure 1: A narrative film with post-war experience. (a) Three identities’ switching starts different narrative. (b) Set interactable objects to make clearly cognize the roles. (c) Continuous subtitles print out the storyline information.

ABSTRACT

“Once” is an interactive virtual reality narration film based on post-war experience, which is set after a continuous war period between two countries. Current virtual reality creations about warfare have only been describing the war itself, and its expression easily causes the audience’s lack of social empathy, which results in weakening the portrayal of anti-war consciousness. More importantly, this kind of creation just narrates in one single view of angle and performs imperfectly. Due to those issues, through the three-identity switching, subtitled guidance and stylized visualization, we create this VR narrative film, which uses the image of common people after the war as clues to express the trauma it brings.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts.

KEYWORDS

Post-war experience, Virtual reality films, Multi-perspective narrative, Subtitle guidance

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1 INTRODUCTION

The two world wars in history allow the will of negating wars and appealing peace lives long in art expression of this century. Y. Wang presents that traditional military-based movie emphasizes the cruelty of the war and accentuates the glory of the winner, but the problem is that over-degrading enemies make the discussion of anti-war lack of accuracy [Oppenheim and Lloyd Okita, 2020][Yulan, 2019] [Todd Ogle et al., 2020]. With the innovation of media, it comes virtual reality creations about warfare theme which focus on shooting games, immersive traveling and panoramic documentaries. Most of them enhance the experience of warfare by rendering the realistic scene of war or making the war process gamified, and those actions easily influence the establishment of social empathy and anti-war consciousness.

As the development of creations, daily-realism-based narrating aspect coming from personal experience is highlighted in anti-war movies which presented by “Schindler’s List”, 1993 and “Life Is Beautiful”, 1997. By the reconstruction of narrative perspective in these anti-war creations, a series of new kinds of war-themed games emerge after 2010, such as “Valiant Hearts: The Great War” and “This Is My War”, which is deeply empathizing players and strengthening the anti-war core.

As a result, we abstract the war process and focus on the three cores of perspective switching, subtitled guidance and interactable plots, to create anti-war related VR films.

2 MULTI-PERSPECTIVE NARRATIVE

Today’s film is based on the time and space elements, through the cross space and multiple time lines, it can carry out two-way communication. In our work, we use three spaces implying different identities to express common persons’ experience. We select doctor, child and reporter as main character identities to describe a same

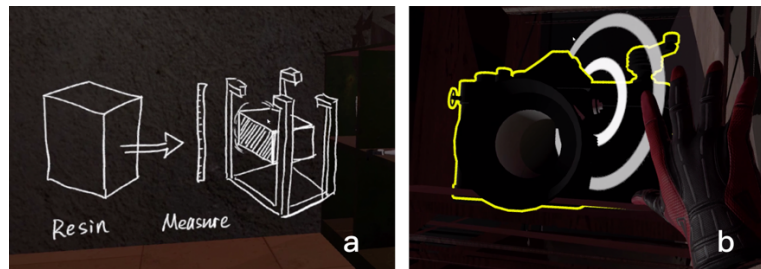


Figure 2: (a) The prosthesis manufacturing process is expressed by the stylized visualization. (b) The symbolic objects are set in scenes.

event at a same time, and set interactable objects in the initial part of the three different spaces to make the audience clearly cognize their roles in time. The situation of the doctors in the prosthetic hospital in the war is repeated scenario, and through interacting with the repeated prosthetic and clinic elements, the patients' stories in the war will be triggered by audience. For children's space, taking advantage of dark environment and by picking up children's daily necessities to guide audience feeling, the audience will see a survival mode of refugee children. And about the journalist's space is based on the disordered sound of broadcast and camera shutter, which expresses the pressure of public opinion outside the battlefield. (Figure 1)

3 SUBTITLE GUIDANCE

In the process of multimodal propagation, the form of text is inefficient for the viewer's cognitive process, also distracts the attention, forming cognitive load. However, it has an advantage from memorized level [S. and James, 2014]. In our work, we integrate three scenes and interactable objects through subtitles appearing. Continuous subtitles will print out the storyline information and point out the interactive elements in the scene, in which the implied text elements can clearly narrate the film plots, and then dissipate automatically after a certain period of time. Meanwhile, the more straightforward subtitle bars in the scene help the viewer pay more attention to the interactable points quickly, which can lead the viewers to the interaction directly and avoid the way loss. Moreover, the design of our three scenes is based on the meaningful elements to guide audience identify three identities, and the use of subtitles can avoid the cognitive obstacles about the elements. (Figure 2)

4 STYLIZED VISUALIZATION

Viewers pay more and more attention to the visual beauty, and they expect the general aesthetic feeling of art works, which makes the whole scene and user interface need stylized design. For post-war scenes, they need to be emphasized the overall dark and broken style. Therefore, we have fragmented all objects in the scenes to avoid the true sense of the scenes appearing and damaging the narrative atmosphere. In different spaces, we set different lighting groups to form different environments, giving psychological hints to the viewer to clarify the identity cognition in the space. At the same time, the fragmentation of the three scenes is also dynamically designed to increase the viewers' interest of scene design.

Meanwhile, to enhance the immersive sense of war, we express the real data from relevant wars to hand-printed visualization in the situations, which improves the viewer's understanding of the war in the context.

5 CONCLUSION

Our film constructs interactive narrative process through multi-perspective narrative, subtitle guidance, and the stylized visualization three aspects to strengthen the narrative in virtual reality film with the theme of post-war experience through three identities. In the future, through these innovative design points, the content under anti-war theme will get more plentiful.

REFERENCES

- David Oppenheim and Randall Lloyd Okita. 2020. The Book of Distance: Personal Storytelling in VR. In *ACM SIGGRAPH 2020 Immersive Pavilion* (SIGGRAPH '20). Association for Computing Machinery, New York, NY, USA, Article 5, 1–2. DOI:https://doi.org/10.1145/3388536.3407896
- Wang Yulan. 2019. Meiguo Junlvitcai Dianying Jiqi Fanzhanyishi Jiexi [Analysis of American Military Theme Movies and Their Anti-War Consciousness]. *Dianying Wenxue* (09), 132–134. doi:CNKI:SUN: dylx.0.2019-09-037.
- Brasel, S. & Gips, James. 2014. Enhancing television advertising: Same-language subtitles can improve brand recall, verbal memory, and behavioral intent. *Journal of the Academy of Marketing Science*. 42. 10.1007/s11747-013-0358-1.
- J. Todd Ogle, Zach Duer, David Hicks, and Scott Fralin. 2020. If this place could talk ... First World War tunnel warfare through haptic VR. In *ACM SIGGRAPH 2020 Immersive Pavilion* (SIGGRAPH '20). Association for Computing Machinery, New York, NY, USA, Article 1, 1–2. DOI:https://doi.org/10.1145/3388536.3407887